Alex Rivera

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Education

Northfield University Toronto, ON

Bachelor of Science in Computer Science, Minor in Statistics *Sep. 2022 – Apr. 2026*

- GPA: 3.8/4.0; Relevant Coursework: Distributed Systems, Databases, Operating Systems, Computer Networks

Experience

Software Engineer Intern May 2025 – Aug. 2025

Lumen Payments – Merchant Platform *Toronto, ON*

- Built a Go payout reconciliation service consuming Kafka ledger events, cutting reconciliation delay from 24 hours to under 5 minutes
- Added idempotency keys and retries to the payout API, taking duplicate payouts from ~12 per month to 0 in the 8 weeks after launch
- Reduced p95 latency of the merchant balance gRPC endpoint from 420ms to 90ms with load tests and PostgreSQL index tuning
- Deployed services to Kubernetes with Helm and added Grafana dashboards and alerts for on-call visibility

Full-Stack Developer Intern May 2024 – Aug. 2024

Brightpath Learning – Student Experience *Remote*

- Designed RESTful APIs in Node.js & Express for assignment submissions, enforcing per-school PostgreSQL row-level security
- Cut image upload failures from 6% to under 0.5% by moving uploads to S3 presigned URLs with a Lambda thumbnail generator
- Raised test coverage from 35% to 78% with Jest and wired the suite into GitHub Actions CI/CD

Teaching Assistant, Data Structures Sep. 2023 – Apr. 2024

Northfield University *Toronto, ON*

- Built a Python autograder with Docker sandboxes that reduced grading time for a team of 4 TAs
- Led weekly tutorials for 35 students on trees, graphs, and hashing in Java

Projects

tinykv | *Go, Raft, gRPC* | github.com/alexrivera/tinykv Oct. 2023 – Dec. 2023

- Implemented Raft leader election and log replication in Go for a sharded key-value store
- Wrote a fault-injection harness that kills and partitions nodes to verify linearizability
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DocuChat | *Python, FastAPI, PostgreSQL, pgvector, Docker, AWS ECS* Jan. 2025 – Present

- Built a retrieval-augmented generation (RAG) app with FastAPI that answers questions over PDFs with cited sources
- Raised answer accuracy on a 50-question eval set from 64% to 82% with hybrid keyword and pgvector search in PostgreSQL

Technical Skills

Languages: Go, Python, TypeScript, Java, SQL, JavaScript, C++, Bash
Frameworks & Libraries: gRPC, Node.js, Express, FastAPI, React, Next.js, Jest, pytest
Cloud, DevOps & Databases: PostgreSQL, Kafka, Kubernetes, Docker, AWS (Lambda, S3, ECS), Terraform, GitHub Actions, Redis
Tools & Platforms: Git, Linux, Grafana, Helm, Claude Code, Cursor